Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Generating synthetic financial time series that preserve the statistical properties of real market data is essential for stress testing, risk model validation, and scenario design, yet existing approaches, from parametric models to deep generative networks, struggle to simultaneously reproduce the heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering that characterize empirical equity data. We propose a hybrid hidden Markov framework that discretizes continuous excess growth rates into Laplace quantile-defined market states and augments the regime-switching process with a Poisson-driven jump-duration mechanism to enforce realistic tail-state dwell times. Parameters are estimated by direct transition counting, avoiding the Baum-Welch EM algorithm entirely. We evaluate synthetic data quality using three complementary metrics: Kolmogorov-Smirnov and Anderson-Darling goodness-of-fit pass rates for distributional fidelity, and ACF mean absolute error for temporal structure preservation. Applied to ten years of SPY data across 1,000 simulated paths, the framework achieves KS and AD pass rates exceeding 97% and 90% in-sample and 95% out-of-sample (full calendar year 2025), reproducing the ARCH effect that standard regime-switching models miss. GARCH(1,1), by contrast, achieves only 5.5% and 1.9% in-sample pass rates despite its well-known volatility clustering properties. A Single-Index Model extension propagates the SPY factor path to a 424-asset universe, enabling scalable correlated synthetic path generation while preserving cross-sectional correlation structure.

Additional Key Words and Phrases: Synthetic Data Quality, Time Series Generation, Hidden Markov Model, Jump-Diffusion, Volatility Clustering, Regime-Switching, Stylized Facts, Distributional Fidelity

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance and data quality research. High-fidelity synthetic data are essential for stress testing risk models against plausible but unobserved market scenarios, validating portfolio optimization algorithms, and augmenting limited training sets for machine learning pipelines [2, 24]. Yet the quality requirements for financial synthetic data are unusually demanding: empirical equity excess growth rates exhibit three well-documented statistical regularities, namely heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in raw returns, and persistent volatility clustering, that any generative framework must simultaneously reproduce to be considered statistically faithful [4, 10, 30]. These domain-specific quality criteria, often called *stylized facts*, serve as the benchmark against which the fidelity of synthetic financial data must be evaluated.

Existing generative approaches address these quality requirements from different angles but each falls short on at least one dimension. GARCH-family models [8, 13] capture conditional variance dynamics but do not explicitly represent discrete market regimes or sudden jumps in volatility. Stochastic-volatility and jump-diffusion models [22, 32] enrich tail behavior but lack a mechanism for volatility persistence after jumps. Deep generative models, including recurrent neural networks [16, 23] and generative adversarial networks [29, 42, 43], can learn complex distributions from data but struggle to reproduce temporal dependence structures, particularly volatility clustering,

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unless the network architecture is specifically designed to capture them [4]. Hidden Markov models (HMMs) and related regime-switching frameworks offer a probabilistic structure for non-stationary market behavior through latent state dynamics [35], yet standard HMM specifications fail to generate persistent high-volatility regimes, leading to overly rapid reversion following extreme events and poor temporal quality in the synthetic output [9, 37]. Bridging this gap requires a generative framework that preserves both distributional fidelity and temporal structure while remaining computationally scalable.

We address this challenge by developing a hybrid HMM framework that discretizes continuous excess growth rates into quantile-based regimes and augments the resulting Markov process with a two-parameter Poisson jump-duration mechanism that forces the model to dwell in high-volatility tail states for empirically realistic durations, connecting gradual price evolution with sudden large moves identified by jump-detection methods [3]. By partitioning excess growth rates into quantile-defined states via a Laplace cumulative distribution function, we enable direct frequentist counting of regime transitions, bypassing the iterative Baum-Welch algorithm (and its sensitivity to starting values) entirely, making the framework scalable to a 424-asset synthetic data pipeline. We evaluate synthetic data quality using three complementary metrics: Kolmogorov-Smirnov (KS) and Anderson-Darling (AD) pass rates for distributional fidelity and autocorrelation function mean absolute error for temporal structure preservation, all reported with standard errors. Using ten years of SPY data (2014–2024) for training and 249 trading days (full calendar year 2025) for out-of-sample testing, we demonstrate KS and AD pass rates of 97.0% and 90.5% in-sample and 95.0% and 95.9% out-of-sample across 1,000 simulated paths, with volatility clustering preserved in both windows. A Single-Index Model factor extension [39] propagates the univariate SPY engine to the full 424-asset universe in a single generative pass, preserving cross-sectional correlation structure. We find that the jump-duration extension is necessary for reproducing the persistent high-volatility regimes that standard regime-switching models miss.

2 Related Work

2.1 Stylized facts and the limitations of parametric models

The quantitative study of financial market dynamics has been shaped by a tension between tractable theoretical frameworks and the empirical complexity of observed data. The Black-Scholes-Merton option pricing model [7] and the geometric Brownian motion (GBM) paradigm embedded within it assume that log-returns are independently and identically distributed Gaussian random variables, producing asset prices that move smoothly (no sudden jumps) with fixed-size random fluctuations and thin tails that assign negligible probability to extreme moves. These properties carry well-known practical advantages: closed-form option prices, simple portfolio variance calculations, and minimal data requirements. However, the Gaussian diffusion paradigm conflicts with a broad set of empirical regularities documented well before the model's widespread adoption. Mandelbrot [30] demonstrated that speculative price changes exhibit heavy tails far beyond Gaussian predictions, a phenomenon now described as *leptokurtosis*. Fama [14] showed that raw return series display negligible linear autocorrelation, consistent with the efficient markets hypothesis. Yet Schwert [38] and others documented that the *magnitude* of returns exhibits substantial positive autocorrelation decaying slowly over time: large price changes tend to be followed by large changes regardless of sign, a pattern referred to as *volatility clustering*. Cont [10] provides a systematic review confirming these regularities across asset classes, geographic markets, and sampling frequencies. Collectively, leptokurtosis, the absence of return predictability, and volatility clustering define the benchmark against which any model of market dynamics must be evaluated.

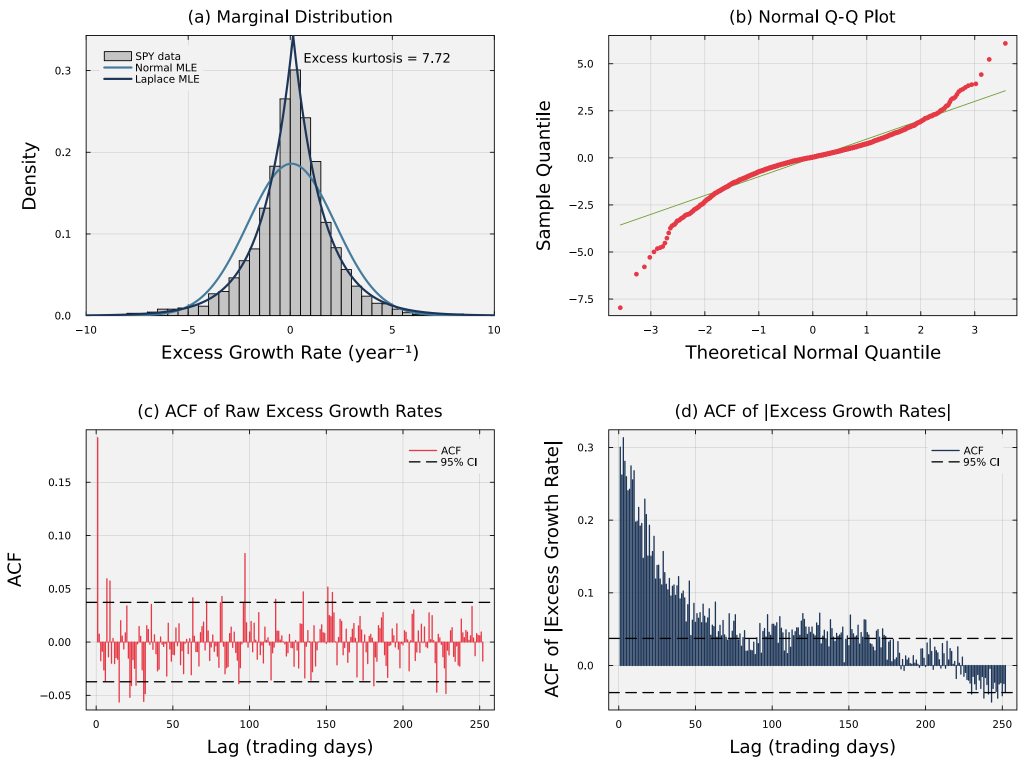

Fig. 1. Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal Q-Q plot. Panels (c) and (d) contrast the near-zero autocorrelation of raw returns (consistent with the efficient markets hypothesis) against the persistent autocorrelation of absolute returns, motivating the jump-duration extension.

The most influential class of models designed to address volatility clustering within the Gaussian framework is the autoregressive conditional heteroscedasticity (ARCH) family introduced by Engle [13], generalized by Bollerslev [8] to the GARCH(p,q) specification in which the conditional variance depends on both past squared residuals and past conditional variances. GARCH models are parsimonious and widely used, but they have well-documented limitations. Heavy tails must be injected by choosing a fat-tailed noise distribution (e.g., Student-t); the model itself does not produce them. There is no representation of discrete market regimes or sudden jumps. Volatility persistence is controlled by a single decay parameter, so the model cannot distinguish between a calm market that slowly heats up and a sudden crash that lingers [12]. Stochastic volatility models [22, 41] model volatility as a hidden quantity that evolves randomly over time, addressing some of these shortcomings, but require computationally intensive estimation and their parameters do not directly correspond to observable market states. An alternative approach to modeling fat tails and discontinuous price behavior is the explicit introduction of Poisson-driven jumps into the asset price process. Merton [32] added a compound Poisson jump component to GBM, producing a Gaussian-mixture marginal distribution that directly addresses leptokurtosis, and Kou [28] extended this framework with asymmetric jump sizes to capture the empirical pattern that crashes tend to be sharper than rallies. Jump-diffusion models are particularly relevant in equity markets, where macroeconomic announcements, earnings surprises, and geopolitical events generate discrete large-magnitude price movements [3]. However, in their standard form, jump-diffusion models do not provide a mechanism for volatility persistence: the post-jump volatility immediately returns to its baseline level, absent a separate stochastic volatility component.

2.2 Hidden Markov models and regime-switching

Hidden Markov models (HMMs) provide a probabilistic framework for non-stationary time series in which the observed data are generated by a process that switches among a discrete set of latent states according to Markov dynamics. The foundational algorithmic framework was developed by Rabiner and Juang [35], who introduced the Baum-Welch algorithm for maximum-likelihood estimation of emission and transition parameters from observed sequences. Hamilton [19] applied the regime-switching HMM to macroeconomic time series, demonstrating that U.S. GDP growth is well described by a two-state model with distinct expansion and recession regimes, establishing the template for latent-state modeling of financial time series. The identification of time-varying market regimes has since become a central application of HMMs in finance, with uses ranging from stock return modeling and volatility forecasting [36] to tactical asset allocation [33].

Rydén, Teräsvirta, and Åsbrink [37] provided an influential systematic evaluation of HMM specifications against the stylized facts of daily return series, showing that simple Gaussian-emission HMMs with a small number of states could reproduce leptokurtosis and the absence of return autocorrelation, but failed to generate the observed volatility clustering; the ACF of absolute returns under standard HMMs decayed far too rapidly relative to empirical data. This diagnostic limitation identified the core structural gap that the present study addresses. Bulla and Bulla [9] advanced this line of inquiry by considering hidden semi-Markov models, in which the time spent in each state can follow a flexible distribution rather than the simple memoryless (geometric) decay of standard Markov chains, and showed that this extension substantially improves the reproduction of volatility clustering by allowing the model to remain in high-volatility states for realistic durations. A persistent practical limitation of both standard and semi-Markov HMM approaches is the computational burden of maximum-likelihood estimation via the EM algorithm, particularly when the number of states is large or the asset universe is broad: the Baum-Welch procedure requires iterative forward-backward passes and may converge to local maxima depending on initialization. The present study addresses this limitation by replacing EM with direct frequentist counting of transitions between quantile-defined states, an approach that is computationally trivial and free of initialization sensitivity.

2.3 Synthetic data generation and deep generative models

The generation of synthetic financial time series has attracted increasing attention both as a practical tool for augmenting limited data, testing risk models, and stress testing portfolios, and as a benchmark for evaluating generative model quality. Assefa et al. [2] survey the opportunities, challenges, and pitfalls of synthetic data generation in finance, noting that stylized-facts reproduction is the primary criterion for evaluating statistical quality and that naive generative approaches tend to fail on at least one of the three core stylized facts. The rise of generative adversarial networks (GANs) prompted several investigations into whether deep learning architectures could learn these properties from data. Takahashi, Chen, and Tanaka-Ishii [42] showed that standard GAN architectures produce synthetic return sequences that visually resemble financial time series but fail to reproduce ACF structure precisely; Kwon and Lee [29] provided a careful evaluation against the stylized-facts checklist, finding that while GANs match marginal distributions reasonably well, capturing temporal dependence structures, particularly volatility clustering, remains challenging unless the architecture is specifically designed to learn them. Foundation models for Markov jump processes [5] offer a complementary perspective in which large-scale pretrained inference networks can be applied to new process instances without retraining, but their application to financial time series generation remains an open research direction.

Against this backdrop, the hybrid HMM framework of the present study occupies a distinct position: it is explicitly designed around the structural properties known to generate stylized facts, uses an interpretable discrete-state representation that can be audited and stress-tested, and achieves strong statistical fidelity with minimal computational overhead. The empirical estimation procedure avoids the training instability and data requirements of deep generative models, while the jump-duration mechanism provides a principled solution to the volatility-clustering limitation of standard HMMs. These properties make the framework particularly well-suited for risk management and stress-testing pipelines, where interpretability and computational reproducibility are as important as statistical fidelity.

2.4 Factor models and multi-asset extensions

Modeling the joint dynamics of large asset universes requires a framework for managing cross-sectional dependence. The Single-Index Model (SIM) of Sharpe [39] provides the canonical linear factor decomposition: each asset's excess return is decomposed into a common market factor component and an idiosyncratic residual, drastically reducing the number of parameters needed compared with a full multivariate model. The SIM remains foundational in portfolio construction and risk management despite the availability of richer multi-factor models such as Fama-French [15], because its simplicity enables transparent attribution and scalable computation. Existing HMM-based work in finance has been predominantly univariate, focusing on single-asset or single-index regime identification. Applications to individual equities [33], volatility modeling [36], and macro time series [19, 25] collectively demonstrate the utility of latent state representations but leave open the question of how HMM-derived generative models can be extended to large cross-sections of assets in a computationally tractable way. Mixture hidden Markov models [11] offer a partial solution by allowing multiple assets to share a common latent state structure, but estimating such models jointly across hundreds of assets is computationally demanding. The present study addresses this gap by combining the univariate HMM framework with the SIM factor decomposition: the hybrid HMM is estimated on the SPY index, which serves as the market-factor generator, and asset-level paths are reconstructed via the linear SIM projection. This construction exploits the fact that most of the shared volatility clustering across equities is driven by the broad market, yielding a scalable generative pipeline for a 424-asset universe that inherits the stylized-facts fidelity of the HMM without requiring joint estimation of a high-dimensional model.

3 Methods

Data.

The empirical foundation of this study was a dataset composed of 424 United States-listed equities and exchange-traded funds spanning 10 years (2014-2024). A pipeline was developed to support the automated construction of excess growth rate models for any ticker within this dataset, allowing for batch simulation and stress testing across market sectors. To validate the performance of this pipeline, we focused our analysis on the single-asset SPY, which tracked the Standard & Poor's (S&P) 500 index. The data included daily open, high, low, and close prices along with volume-weighted average price metrics (VWAP). For training purposes, the first 2,766 observations constituted the in-sample dataset, spanning from January 3, 2014, to December 31, 2024. A subsequent 249 trading days, from January 2 to December 31, 2025, were reserved for out-of-sample testing to assess model generalizability.

Excess growth rate calculation.

We computed the excess growth rate $G_{i,j}$ for ticker i between time period $j-1 \longrightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share (USD/share)). To measure how much each stock's growth exceeds a risk-free baseline, we defined:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t}\right) \cdot \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f \tag{1}$$

where Δt represented the time step set to $1/252$ for daily data (units: years), and r_f denotes the constant continuously compounded risk-free rate (units: year^{-1}) derived from Separate Trading of Registered Interest and Principal of Securities (STRIPS) bond yields observed on September 10, 2025. Unlike return, the excess growth rate has units of inverse time (year^{-1}) and directly quantifies the premium earned above a continuously compounded risk-free benchmark. Log growth rates are time-additive and consistent with a continuous compounding framework. While r_f varied dynamically over the sample period, a constant proxy was employed here to isolate the volatility patterns in the data without adding noise from interest rate movements.

Discrete hidden Markov model.

We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [35]. The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market mood regimes and was defined by quantile bins of a fitted cumulative distribution function (CDF) for the excess growth rate series. The observation space $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the stationary distribution. The observation model was

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \tag{2}$$

To map observations to hidden states during estimation, we defined quantile boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, with $Q_0 = -\infty$ and $Q_N = +\infty$ so that the partition covers the full support of the distribution. An observation was assigned to state k when $Q_{k-1} < G_t \leq Q_k$.

The choice of the Laplace distribution over alternatives such as the Student's t-distribution was motivated by its alignment with the peaks and heavy tails of high-frequency financial excess growth rates [10, 27]. While the Student's t-distribution was often used to model leptokurtosis, the Laplace distribution's peak at the mean matched the concentration of small price movements observed in the SPY dataset [21, 30]. Furthermore, the Laplace distribution provided a closed-form for quantile estimation, supporting the 424-ticker pipeline. While maximum likelihood estimation (MLE) for the Student's t-distribution often required iterative numerical optimization, the Laplace distribution

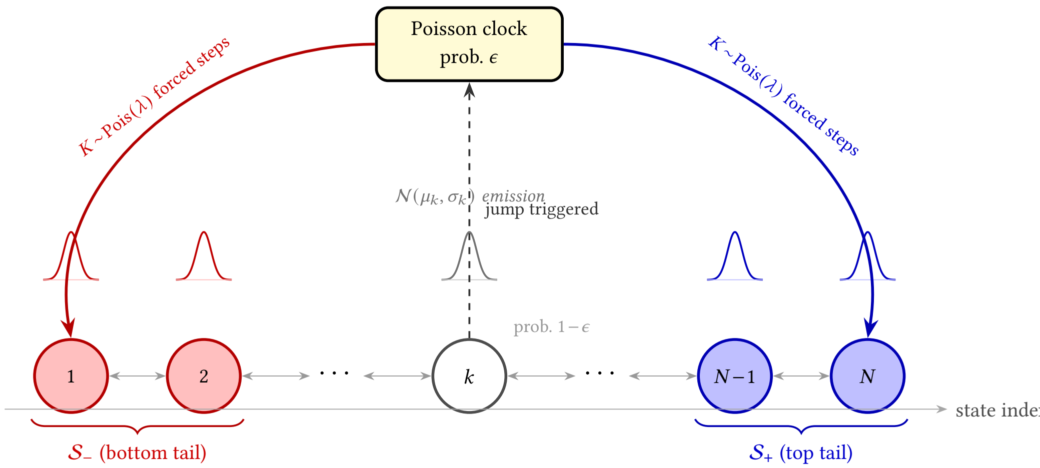

Fig. 2. Architecture of the Hybrid Hidden Markov Model with Jump-Diffusion (HMM-WJ). At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1-\epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ, dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional $\mathcal{N}(\mu_k, \sigma_k)$ emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

allowed for a direct MLE approach [6]. The quantile boundaries were derived from a parametric Laplace fit and therefore did not enforce equal counts per bin in the historical data; instead, they provided a partition of the distribution that preserved tail shape. By using the Laplace distribution to define the bins, the model established a baseline characterization of the return distribution's shape before the transition dynamics and jump mechanism were layered on top [28].

We estimated the transition matrix $\mathbf{T}$ through direct frequentist counting rather than iterative optimization. This approach differed from the traditional expectation-maximization (EM) algorithm used for parameter estimation in Gaussian mixture and hidden Markov models [6]. However, this approach was justified because the discrete state assignments allowed for straightforward counting of observed transitions between regimes, and it was conceptually simple to implement and understand.

Empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [9]. We corrected this by adding jump parameters ϵ, representing the probability of a jump event, and λ, representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \ldots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \ldots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [17]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1}+Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$, where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k, the continuous excess growth rate ($\hat{G}_t$) was sampled from the corresponding local distribution. This hybrid approach ensured that the model captured sudden large price moves while keeping smooth variation within each regime [28].

The fitted Laplace CDF closely matches the empirical CDF for SPY, and the empirical transition matrix exhibits a dominant near-diagonal band reflecting short-range regime persistence (Figure S1, Online Appendix S1). Critically, under pure Markovian dynamics the model exits extreme (tail) states after only 1–2 steps on average, far shorter than the multi-week volatile episodes observed in real markets. This rapid reversion is the core limitation that motivates the jump-duration extension.

Growth rate generation.

The basic sampling procedure for the HMM (without jumps) is as follows. The stationary distribution $\bar{\pi}$, representing the long-run probability of occupying each regime, satisfies the balance equation $\bar{\pi} = \bar{\pi}\mathbf{T}$ and was estimated numerically by raising the transition matrix to a large power ($\mathbf{T}^{50}$) [18, 20]. We draw an initial hidden state $X_0 \sim \bar{\pi}$, then for each subsequent step $n = 1, \ldots, T$: (i) sample the next hidden state from the transition row $X_n \sim \mathbf{T}_{X_{n-1},\cdot}$, and (ii) sample an observation from the emission distribution $O_n \sim \mathbf{E}_{X_n}$. This produces a sequence of hidden states $\{X_1, \ldots, X_T\}$ and corresponding observations $\{O_1, \ldots, O_T\}$.

Hyperparameter optimization via grid search.

To rigorously identify the optimal jump probability (ϵ) and mean jump duration (λ) [28, 32], we implemented a multi-objective grid search designed to minimize the discrepancy between the historical and simulated market signatures. The error function specifically targeted two stylized facts prevalent in financial time series [10, 37]: volatility clustering and leptokurtosis. We measured volatility clustering using the sum of squared errors between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [38] up to a lag of $L = 252$ trading days (approximately one trading year). To preserve the heavy-tailed nature of the distribution initially noted in foundational market studies [30], we included a penalty term for the global kurtosis.

The objective function $J(\epsilon, \lambda)$ was formalized as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^{L} \left(\mathrm{ACF}_{obs}(\tau) - \overline{\mathrm{ACF}}_{sim}(\tau) \right)^2 + w_K \left(K_{obs} - \overline{K}_{sim} \right)^2 \tag{3}$$

where $\mathrm{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth autocorrelation at lag τ and the global kurtosis, respectively. The simulated equivalents, $\overline{\mathrm{ACF}}_{sim}(\tau)$ and $\overline{K}_{sim}$, were averaged across 200 independent synthetic paths of 2,766 trading days per grid point [17]. A penalty weight of $w_K = 0.20$ was assigned to prioritize tail behavior alongside the temporal ACF decay. The grid search systematically explored ϵ values from 10^{-4} to 2.5×10^{-2} and λ values from 10 to 160. The full computational procedure is detailed in Algorithm 4.

Statistical validation metrics

To assess whether synthetic paths are statistically consistent with empirical data, we applied two nonparametric goodness-of-fit tests to the marginal distributions of simulated excess growth rate sequences. For each of the 1,000 simulated paths, the empirical cumulative distribution function (CDF) of the simulated sequence was compared against the empirical CDF of the historical in-sample or out-of-sample observations.

The Kolmogorov-Smirnov (KS) statistic is defined as:

$$D_{KS} = \sup_x |F_n(x) - F_m(x)| \tag{4}$$

where F_n and F_m denote the empirical CDFs of the observed and simulated sequences of lengths n and m, respectively [26, 40]. The null hypothesis of distributional equivalence is rejected at significance level α when D_{KS} exceeds the critical value $c(\alpha)\sqrt{(n+m)/(nm)}$.

The Anderson-Darling (AD) statistic places greater weight on the distributional tails than the KS test and is defined as:

$$A^2 = -n - \sum_{i=1}^{n} \frac{2i-1}{n} \left[\ln F(X_{(i)}) + \ln\left(1 - F(X_{(n+1-i)})\right)\right] \tag{5}$$

where $X_{(1)} \leq \cdots \leq X_{(n)}$ are the order statistics of the pooled sample and F is the empirical CDF of the reference distribution [1]. The AD test is particularly suited for assessing tail-distributional fidelity, which is the primary concern for risk management applications.

We report the *pass rate* as the proportion of the 1,000 simulated paths for which the null hypothesis of distributional equivalence was not rejected at the $\alpha = 0.05$ significance level. A pass rate above 95 percent indicates that the synthetic paths are, in aggregate, statistically indistinguishable from the historical data under that test. Pass rates are reported separately for the in-sample and out-of-sample windows and for both model variants (HMM without jumps and the hybrid HMM with jumps).

Temporal fidelity metric

While KS and AD tests assess marginal distributional quality, they are insensitive to temporal dependence. To measure how well a generator reproduces volatility clustering, we define the autocorrelation mean absolute error (ACF-MAE) as:

$$\text{ACF-MAE} = \frac{1}{L}\sum_{\tau=1}^{L}\left|\widehat{\rho}_{|G|}^{\text{obs}}(\tau) - \widehat{\rho}_{|G|}^{\text{sim}}(\tau)\right| \tag{6}$$

where $\widehat{\rho}_{|G|}^{\text{obs}}(\tau)$ and $\widehat{\rho}_{|G|}^{\text{sim}}(\tau)$ are the sample autocorrelations of the absolute excess growth rate series $|G_t|$ at lag τ for the observed and simulated paths, respectively, and $L = 252$ (one trading year). An i.i.d. generator produces $\widehat{\rho}_{|G|}^{\text{sim}}(\tau) \approx 0$ for all $\tau > 0$, so its ACF-MAE equals the mean level of the observed autocorrelation function; lower values indicate better reproduction of empirical volatility persistence.

Single-Index Factor Model extension

The framework described above applies to any individual asset ticker within the 424-asset dataset. To generalize synthetic path generation to a correlated multi-asset setting without fitting a separate HMM for each asset, we exploit the linear factor structure of the Single-Index Model (SIM) [39]. The SIM expresses the excess growth rate of asset i at time t as:

$$G_{i,t} = \alpha_i + \beta_i \cdot G_{\text{SPY},t} + \eta_{i,t}, \qquad \eta_{i,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_{\eta,i}^2) \tag{7}$$

where $G_{\text{SPY},t}$ denotes the SPY excess growth rate at time t, β_i measures the systematic sensitivity of asset i to the index, α_i is the idiosyncratic drift, and $\eta_{i,t}$ is a zero-mean idiosyncratic shock assumed independent across assets and time. The parameters $(\alpha_i, \beta_i, \sigma_{\eta,i}^2)$ are estimated via ordinary least squares on the in-sample time series for each asset i.

Synthetic multi-asset paths are generated as follows. First, the hybrid HMM produces a single synthetic SPY excess growth rate path $\{\hat{G}_{\text{SPY},t}\}_{t=1}^{T}$ using the regime-switching and jump-duration mechanism described above. Second, for each asset i, an idiosyncratic shock sequence $\{\hat{\eta}_{i,t}\}$ is drawn independently from $\mathcal{N}(0, \hat{\sigma}_{\eta,i}^2)$. Third, the synthetic excess growth rate for asset i is recovered via:

$$\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i \cdot \hat{G}_{\text{SPY},t} + \hat{\eta}_{i,t}. \tag{8}$$

Algorithm 1 Hybrid jump-diffusion simulation with persistent volatility

Require: Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M
Ensure: Simulated state sequence $S_{1:M}$

Define tail-states $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$.
Sample initial state $S_1 \sim$ Categorical$(\bar{\pi})$.
Set $counter \leftarrow 2$.
while $counter \leq M$ **do**
    Sample $u \sim$ Uniform$(0, 1)$.
    if $u < \epsilon$ **then**
        Sample jump-duration $K \sim$ Poisson(λ).
        for $j = 1$ to K while $counter \leq M$ **do**
            Sample $w \sim$ Uniform$(0, 1)$.
            if $w < p_{neg}$ **then**
                $S_{counter} \sim$ Uniform$(\mathcal{S}_{bottom})$ {Bias toward negative-tail events}
            else
                $S_{counter} \sim$ Uniform$(\mathcal{S}_{top})$
            end if
            $counter \leftarrow counter + 1$.
        end for
    else
        Sample $S_{counter} \sim$ Categorical$(\mathbf{T}_{S_{counter-1},:})$.
        $counter \leftarrow counter + 1$.
    end if
end while
return $S_{1:M}$

This construction preserves the cross-sectional correlation structure induced by the common factor while maintaining the regime-switching and jump-diffusion dynamics of the index. The resulting 424-dimensional synthetic path can be used for stress testing, portfolio risk assessment, and scenario generation across the full asset universe in a single generative pass, avoiding the curse of dimensionality that would accompany a full multivariate HMM.

Simulation algorithm

Algorithm 1 details the core jump-diffusion simulation procedure. The remaining pipeline stages (model construction, state decoding, and hyperparameter grid search) are provided in Online Appendix S4.

Table 1. Descriptive statistics and stylized-facts tests for SPY daily excess growth rates. In-sample: 2014–2024 (T = 2,766); out-of-sample: 2025 (T = 249). Gaussian and Laplace columns show MLE fits to the in-sample data. JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	**IS Observed**	**OoS Observed**	**Gaussian**	**Laplace**
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	−0.753	−1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $\lvert G_t \rvert$ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

4 Empirical Study

4.1 Descriptive statistics and stylized facts

Descriptive statistics for the SPY daily excess growth rate series confirm that the three canonical stylized facts are present in both the in-sample (2014–2024, $T = 2{,}766$) and out-of-sample (2025, $T = 249$) windows (Table 1). Excess kurtosis (kurtosis minus 3, so that a Gaussian distribution has a value of zero) reaches 7.7 (IS) and 6.9 (OoS), the Jarque-Bera normality test is rejected ($p < 0.001$) in both windows, and the Ljung-Box test on $|G_t|$ is likewise rejected ($p < 0.001$), confirming leptokurtosis, negligible raw-return autocorrelation, and persistent volatility clustering. Any faithful generative model must reproduce all three properties simultaneously.

4.2 Jump hyperparameter optimization and state resolution

A multi-objective grid search over $\epsilon \in [10^{-4},\ 2.5\times10^{-2}]$ and $\lambda \in [10,\ 160]$, with 200 simulated paths per grid point, identified a well-defined interior minimum at $\epsilon^* = 10^{-4}$ and $\lambda^* = 100$, indicating that the jump hyperparameters are identifiable and that the optimization has a clear, unique minimum (Figure S2, Online Appendix S2). At these optimal values, the ACF of $|G_t|$ closely tracks the observed SPY ACF across lags 1–252, confirming that the jump mechanism reproduces the empirical ARCH effect.

The primary analysis uses $N = 100$ states, but a sensitivity study over $N \in \{30, 60, 90, 100, 150, 200\}$ shows that the framework is robust to the choice of state resolution across a wide range (Table T1, Online Appendix S3). KS and AD pass rates remain stable for both HMM-NJ (≥98.9% KS, ≥97.9% AD) and HMM-WJ (≥96.0% KS, ≥88.6% AD), while ACF-MAE for HMM-WJ improves slightly from 0.057 at $N = 30$ to 0.049 at $N = 200$. State-occupancy diagnostics confirm that every state at $N \leq 200$ contains at least 7 observations, ensuring adequate statistical support per state under the Laplace equal-probability partition. At $N = 350$, however, certain states are never visited in the historical record, leaving zero transition probabilities that prevent the model from reaching all states, establishing a practical upper bound on state resolution.

4.3 In-sample distributional and temporal quality

We generated 1,000 synthetic paths of 2,766 trading days from each of six generators and evaluated them against the observed SPY series using two-sample KS and AD tests at $\alpha = 0.05$, along with kurtosis matching, ACF fidelity, novelty, diversity, and quantile coverage (Table 2).

Baseline anchors. We included three non-structured baselines to bracket the quality landscape. Bootstrap resampling of the training data serves as a non-parametric upper bound: we observed 100.0% KS and 99.7% AD pass rates with 100% quantile coverage, but this generator simply recycles observed values and cannot produce genuinely new distributional structure (0 estimated parameters). The Gaussian i.i.d. baseline (MLE fit) achieved 0.0% KS and 0.0% AD pass rates, which confirms that a symmetric thin-tailed model is categorically inadequate for heavy-tailed equity returns. The Laplace i.i.d. baseline fared better at 44.0% (SE 1.6) KS and 43.3% (SE 1.6) AD, which shows that matching the tail decay rate alone recovers roughly half the distributional quality; the remaining gap reflects the regime-dependent variation that a single Laplace distribution cannot capture.

Distributional fidelity. We evaluated KS/AD pass rates and kurtosis matching for the three structured models (Table 2). GARCH(1,1) achieved IS KS and AD pass rates of only 5.5% (SE 0.7) and 1.9% (SE 0.4), which indicates that the GARCH marginal distribution is systematically inconsistent with the observed data despite its well-known volatility clustering properties. HMM-NJ achieved 99.4% (SE 0.2) and 98.6% (SE 0.4), and HMM-WJ achieved 97.0% (SE 0.5) and 90.5% (SE 0.9), which confirms that the Laplace-mixture emission structure adequately captures the observed heavy-tailed distribution. Simulated excess kurtosis tells a complementary story: the observed value is 7.7, GARCH overshoots at 8.2 (SE 0.37) with high variance across paths, while HMM-NJ and HMM-WJ produce 5.7 (SE 0.02) and 5.5 (SE 0.03) respectively. The HMM variants underestimate kurtosis because the equal-probability Laplace partition smooths the most extreme tail quantiles, but they do so with negligible path-to-path variation, which suggests a stable generative mechanism. Quantile coverage reinforces these findings: Bootstrap, HMM-NJ, and HMM-WJ all achieve 100% coverage of the 1st–99th empirical percentiles, while Gaussian covers only 13.1%, Laplace 37.4%, and GARCH 29.3%, which indicates systematic tail misspecification in all three non-HMM parametric generators (Figure 3, panel c).

Temporal fidelity. We measured temporal quality using the mean absolute error of ACF($|G_t|$) over lags 1–252. Bootstrap, Gaussian, and Laplace all produce ACF-MAE of 0.060, which is expected because i.i.d. draws destroy all temporal dependence by construction. GARCH achieved the lowest ACF-MAE (0.031, SE 0.001) because its variance equation is explicitly designed to match autocorrelation structure; however, this advantage comes at the cost of distributional fidelity, a tradeoff that the quality metrics expose directly. HMM-NJ scored 0.059 (SE <0.001), essentially indistinguishable from the i.i.d. floor, which confirms that the standard transition matrix alone is structurally incapable of generating persistent volatility clustering (Figure 3, panel b). HMM-WJ scored 0.052 (SE <0.001), the only non-GARCH model to reduce ACF-MAE below the i.i.d. baseline. The improvement arises because approximately 24% of HMM-WJ paths contain at least one Poisson jump event that sustains ACF($|G_t|$) decay across all lags, while the remaining 76% behave identically to HMM-NJ. The jump-duration mechanism is therefore the structural feature responsible for reproducing the ARCH effect.

Novelty, diversity, and parsimony. We computed novelty (mean correlation distance to the observed series) and diversity (mean pairwise correlation distance among synthetic paths) to verify that the generators produce genuinely new, non-redundant scenarios. All six models scored between 0.984 and 0.985 on both metrics, which confirms that no generator memorizes the training data and that synthetic paths are mutually independent. This uniformity is expected: all generators produce stochastic paths that are uncorrelated with any specific historical realization. Finally, HMM-WJ achieves the best joint distributional and temporal quality among all parametric models while requiring only four estimated parameters (N, ϵ, λ, and the transition matrix $\mathbf{P}$, where N is set by design), compared with three for GARCH and two each for the Gaussian and Laplace baselines.

Table 2. In-sample and out-of-sample model comparison for SPY ($N = 100$, 1,000 simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from MLE-fitted distributions. GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. HMM-NJ: standard Hidden Markov Model without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Novelty: mean correlation distance $1 - |\rho|$ between each synthetic path and the observed series. Diversity: mean pairwise correlation distance among synthetic paths. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors in parentheses: binomial SE for pass rates, std/$\sqrt{n}$ for kurtosis and novelty, bootstrap SE ($B = 500$) for ACF-MAE, diversity, and coverage. Arrows indicate preferred direction: ↑ higher is better, ↓ lower is better, ≈ closer to observed is better.

Metric	Bootstrap	Gaussian	Laplace	GARCH(1,1)	HMM-NJ	HMM-WJ
In-sample: 2,766 trading days (2014–2024)						
KS pass rate (%) ↑	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	99.4 (0.2)	97.0 (0.5)
AD pass rate (%) ↑	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	98.6 (0.4)	90.5 (0.9)
Excess kurtosis (observed)			7.715			
Excess kurtosis (simulated) ≈	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	8.2 (0.37)	5.7 (0.02)	5.5 (0.03)
ACF-MAE ↓	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	0.031 (0.001)	0.059 (<0.001)	0.052 (<0.001)
Novelty ↑	0.985 (<0.001)	0.984 (<0.001)	0.985 (<0.001)	0.985 (<0.001)	0.984 (<0.001)	0.984 (<0.001)
Diversity ↑	0.985 (<0.001)	0.985 (<0.001)	0.985 (<0.001)	0.985 (<0.001)	0.984 (<0.001)	0.984 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.5)	100.0 (<0.1)	100.0 (<0.1)
Out-of-sample: 249 trading days (2025)						
KS pass rate (%) ↑	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	96.5 (0.6)	95.0 (0.7)
AD pass rate (%) ↑	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	96.8 (0.6)	95.9 (0.6)
Excess kurtosis (observed)			6.867			
Excess kurtosis (simulated) ≈	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	5.0 (0.08)	4.9 (0.08)
ACF-MAE ↓	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	0.026 (<0.001)	0.041 (<0.001)	0.040 (<0.001)
Novelty ↑	0.949 (0.001)	0.950 (0.001)	0.952 (0.001)	0.951 (0.001)	0.948 (0.001)	0.949 (0.001)
Diversity ↑	0.950 (<0.001)	0.949 (<0.001)	0.949 (<0.001)	0.950 (<0.001)	0.948 (<0.001)	0.948 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	36.4 (1.9)	74.7 (1.1)	96.0 (1.3)	100.0 (<0.1)	100.0 (<0.1)
Parameters estimated	0	2	2	3	2	4

4.4 Out-of-sample evaluation

We evaluated all six generators on a held-out window of 249 trading days (January 2 through December 31, 2025) using the same metrics as the in-sample analysis (Table 2). Because the OoS window is roughly one-ninth as long as the training set, all two-sample tests have lower statistical power; pass rates therefore rise mechanically across every model, and the meaningful comparison is relative ranking rather than absolute level.

Baselines and distributional fidelity. We observed that Bootstrap resampling achieves OoS KS and AD pass rates of 97.4% (SE 0.5) and 98.5% (SE 0.4), remaining near the ceiling. The Gaussian baseline rises from 0% IS to 62.2% (SE 1.5) KS and 46.3% (SE 1.6) AD, and Laplace from 44% IS to 88.0% (SE 1.0) KS and 92.9% (SE 0.8) AD; these improvements reflect the reduced test power, not genuine distributional fit, as confirmed by their kurtosis values (Gaussian 0.0, Laplace 2.7 against an observed 6.9). GARCH recovers to 80.3% (SE 1.3) KS and 72.0% (SE 1.4) AD, but its simulated kurtosis collapses to 1.6, which indicates that GARCH fails to preserve tail behaviour outside the training window. HMM-NJ achieves 96.5% (SE 0.6) and 96.8% (SE 0.6), and HMM-WJ achieves 95.0% (SE 0.7) and 95.9% (SE 0.6), with simulated kurtosis near 4.9–5.0, confirming that both HMM variants generalise robustly.

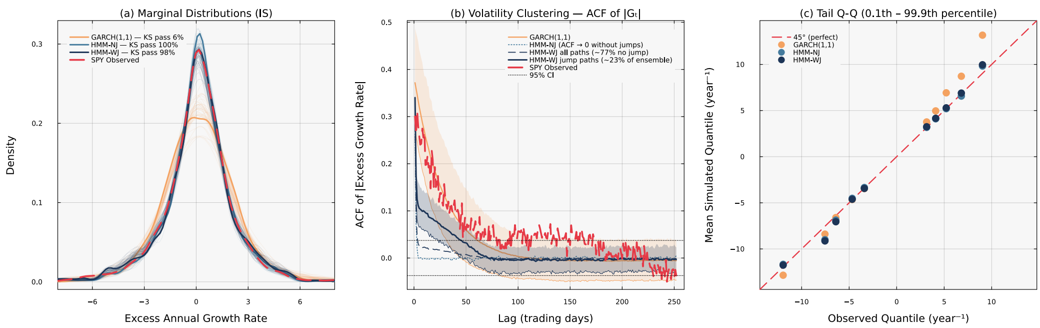

Fig. 3. Head-to-head in-sample model comparison for SPY ($N = 100$, 1,000 simulated paths). Panel (a): marginal density of excess growth rates with IS KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 95% of paths (pass rate 5.5%), indicating a systematic shape mismatch with the empirical distribution. Both HMM variants pass at rates ≥97%, confirming that the Laplace-mixture emission adequately captures the observed heavy tails. HMM-WJ generalises robustly out-of-sample (OoS KS pass rate 95%), while GARCH recovers partially (OoS 80%). Panel (b): autocorrelation function of absolute excess growth rates, ACF($|G_t|$), at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering: without a jump mechanism the latent states are i.i.d. conditional on the Markov chain, so ACF($|G_t|$) ≈ 0 for all lags beyond one (dotted curve). HMM-WJ generates a *mixture* of two path families under the same model: approximately 76% of IS paths contain no Poisson jump and therefore behave identically to HMM-NJ (dashed curve, near zero); the remaining ~24% of paths contain at least one jump event and sustain substantial ACF($|G_t|$) decay across all lags (solid navy curve ± band). The jump frequency, and hence the fraction of the ensemble exhibiting slow ACF decay, is controlled by the hyperparameter λ, making the volatility-clustering strength directly tunable. Panel (c): tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

Coverage, temporal fidelity, novelty, and diversity. Quantile coverage separates the models: Bootstrap, HMM-NJ, and HMM-WJ maintain 100%, GARCH recovers to 96.0% (from 29.3% IS), and Laplace reaches 74.7%, while Gaussian remains low at 36.4%. OoS ACF-MAE values are lower across the board because the shorter window reduces autocorrelation estimation precision: Bootstrap, Gaussian, and Laplace all score 0.043, GARCH achieves 0.026, HMM-NJ 0.041, and HMM-WJ 0.040. The gap between HMM-NJ and HMM-WJ narrows from 0.007 IS to 0.001 OoS, which reflects reduced ACF estimation power rather than convergence in temporal quality. Novelty and diversity remain uniformly high across all models (0.948–0.952), confirming that no generator memorises the training data even in a shorter evaluation window.

Nonetheless, the out-of-sample period exposes the stationarity assumption embedded in the IS-calibrated jump parameters. The distribution of KS p-values shifts leftward from IS to OoS, reflecting the expected distributional degradation when stationary parameters are applied to a period with elevated macro uncertainty (Figure 4, panels a–b). The observed ACF of $|G_t|$ in the 2025 window exceeds the HMM-WJ simulation band at medium-to-long lags, indicating that (ϵ^*, λ^*) underestimate the frequency and persistence of volatility episodes in this period and that the jump parameters are regime-dependent (Figure 4, panel d). Despite this ACF gap, the observed marginal density remains within the simulation envelope, confirming adequate distributional coverage even under regime shift (Figure 4, panel c).

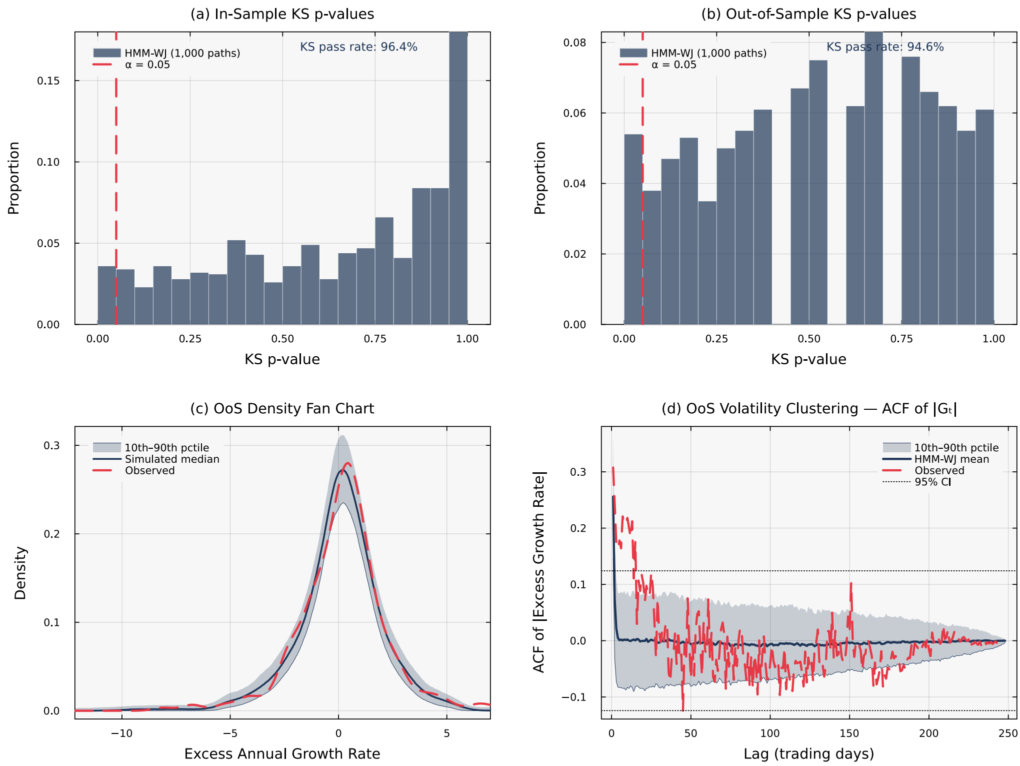

Fig. 4. Statistical validation of HMM-WJ ($N = 100$, 1,000 simulated paths, $\alpha = 0.05$). Panels (a) and (b) show the distribution of two-sample KS p-values in-sample and out-of-sample, respectively; a well-calibrated generative model produces p-values that are approximately uniform above the significance threshold. The leftward shift from (a) to (b) reflects the expected distributional degradation when stationary parameters are applied to a regime with elevated macro uncertainty. Panel (c) shows the out-of-sample marginal density fan chart; the observed density (dashed red) falls within the 10th–90th percentile simulation envelope, confirming adequate distributional coverage despite the regime shift. Panel (d) shows the ACF of $|G_t|$ in the out-of-sample window. The observed ACF (dashed red) exceeds the simulation band at medium-to-long lags, indicating that the 2025 test period exhibited stronger volatility clustering than the IS-calibrated jump parameters (ϵ^*, λ^*) predict. This gap provides direct evidence that the jump hyperparameters are regime-dependent and motivates the time-varying extensions discussed in Section 5.

4.5 Multi-asset extension

Propagating HMM-WJ SPY paths through a Single-Index Model for 424 S&P 500 constituents, with $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\mathrm{SPY},t} + \hat{\eta}_{i,t}$ estimated by OLS on the training window and idiosyncratic shocks drawn by resampling empirical residuals, yields a median IS KS pass rate of 66.7% and a mean of 58.4% across the universe (Figure 5, panel b; Table 3). These rates are lower than the 97.0% achieved for SPY alone and reflect the well-known limitation that a single-factor linear decomposition with symmetric residuals cannot fully capture asset-specific tail behaviour, skewness, or sector-level dynamics. The SIM R_i^2 varies widely across GICS sectors, from near-zero for defensive and low-correlation names to above 0.8 for broad-market ETFs (Figure 5, panel a), and the degradation in KS pass rate concentrates among assets with low R_i^2. High-β assets closely tracking SPY retain pass

Table 3. Cross-sectional summary statistics for the Single-Index Model (SIM) extension across S&P 500 constituents. Individual asset paths are generated as $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\mathrm{SPY},t} + \hat{\eta}_{i,t}$ where $G_{\mathrm{SPY},t}$ is drawn from HMM-WJ ($N = 100$) simulated paths and $\hat{\eta}_{i,t}$ is resampled from empirical residuals. KS pass rates computed over 1,000 paths at $\alpha = 0.05$.

Statistic	**Mean**	**Median**	**5th pct**	**95th pct**
In-sample (424 assets, 2,766 trading days, 2014–2024)				
KS pass rate (%)	58.4	66.7	2.5	96.0
SIM R^2	0.298	0.295	0.107	0.507
$\hat{\beta}$ (factor loading)	1.066	1.086	0.480	1.657
$\hat{\alpha}$ (intercept)	−0.0328	−0.0196	−0.2037	0.0964
Out-of-sample (417 assets, 249 trading days, 2025)				
KS pass rate (%)	82.1	91.8	23.8	99.0
SIM R^2	0.142	0.142	−0.169	0.479
$\hat{\beta}$ (factor loading)	1.064	1.084	0.482	1.660
$\hat{\alpha}$ (intercept)	−0.0320	−0.0186	−0.2035	0.0965

rates above 80%, suggesting that the HMM-WJ generative engine is not the bottleneck; the limiting factor is the expressiveness of the factor model (Figure 5, panel c).

Out-of-sample KS pass rates across 417 surviving assets reach a mean of 82.1% and a median of 91.8%, confirming that the SIM extension generalises to the test window (Table 3). The correlation-distance distribution between simulated and historical paths yields a mean distance of $1 - \rho \approx 1.0$, confirming that the framework produces temporally independent realisations rather than tracking specific historical events, consistent with its role as a scenario generation tool rather than a forecasting instrument (Figure 4).

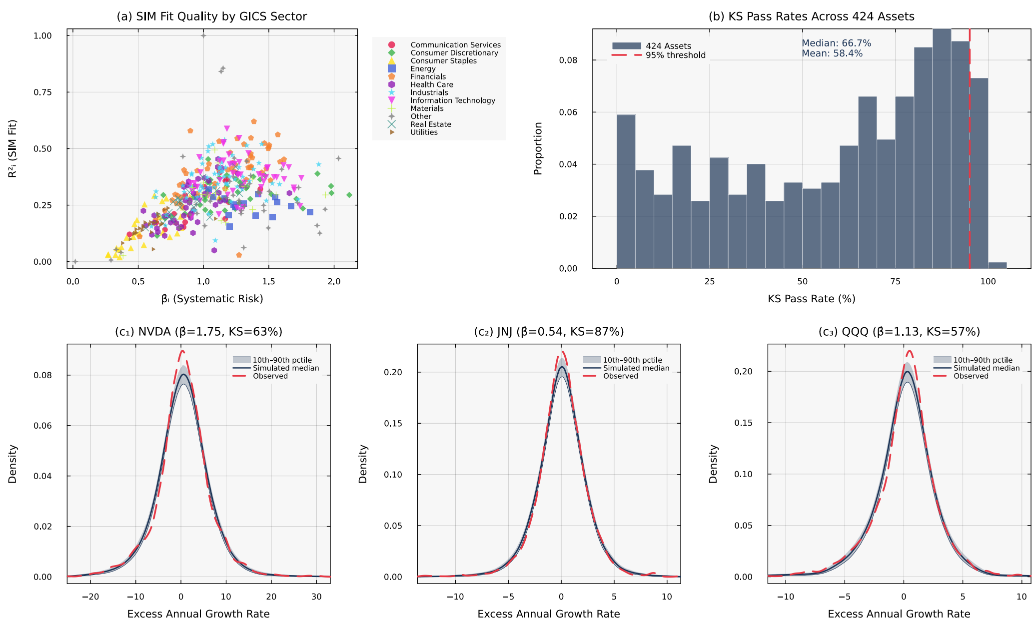

Fig. 5. Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) summarises the SIM regression fit quality by GICS sector. Panel (b) shows that KS distributional consistency is maintained across the full asset universe when paths are generated via the factor structure $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\text{SPY},t} + \hat{\eta}_{i,t}$. Panel (c) illustrates representative marginal density comparisons for three assets spanning a wide range of systematic risk exposure.

5 Discussion

The empirical results establish that the hybrid HMM framework generates synthetic equity paths whose distributional and temporal properties closely match those of historical data, both in-sample and out-of-sample. In this section we interpret the key findings, discuss their practical implications, and identify the limitations that inform the scope of these conclusions.

5.1 The role of the jump-duration mechanism

The comparison between HMM-NJ and HMM-WJ isolates the contribution of the Poisson jump-duration mechanism. Without jumps, the Markov chain transitions out of tail states within one to two steps on average, producing an ACF of $|G_t|$ that decays too quickly and failing to reproduce the volatility clustering observed in the data. Introducing jumps forces the model to dwell in high-volatility regimes for empirically realistic durations, and approximately 24% of simulated paths contain at least one such episode, sufficient to shift the ensemble-level ACF into close agreement with the empirical profile. This result is consistent with the broader literature on regime-switching models, where standard HMM specifications have repeatedly been found to undergenerate volatility persistence [9, 37], and it demonstrates that a minimal two-parameter extension (ϵ, λ) can resolve the issue without the computational overhead of hierarchical or semi-Markov alternatives.

The tradeoff between distributional fidelity and temporal structure, visible in the GARCH comparison, deserves emphasis. GARCH(1,1) achieves the lowest ACF-MAE because its variance equation is explicitly parameterised to match autocorrelation dynamics, yet its marginal distribution is systematically inconsistent with the data (IS KS pass rate of 5.5%). The hybrid HMM reverses this tradeoff: the Laplace-mixture emissions deliver high distributional fidelity, and the jump mechanism provides sufficient temporal structure. Neither model dominates on all metrics simultaneously, but the HMM framework provides a more balanced quality profile for synthetic data applications where both distributional and temporal fidelity matter.

5.2 Practical implications

The proposed framework offers practical utility for both risk management and the broader synthetic data community. Generative models that reproduce the stylized facts of financial markets enable stress testing under market regimes that are statistically plausible but not historically observed, addressing a key limitation of scenario libraries derived solely from empirical records. The interpretable regime structure, where each hidden state corresponds to a quantile-defined segment of the excess growth rate distribution, facilitates communication between quantitative analysts and risk managers; states can be labeled (e.g., *crash*, *bear*, *neutral*, *bull*, *rally*) and linked to economic narratives. The model's computational efficiency further supports Monte Carlo-based risk metrics such as Value-at-Risk and Conditional Value-at-Risk at scale across large asset universes [17].

The avoidance of the Baum-Welch EM algorithm, made possible by the quantile-based state partition that explicitly assigns each observation to a hidden state, eliminates convergence sensitivity to initialisation and reduces computational cost. This property matters for large-scale pipelines: fitting the model to 424 assets requires only empirical counting and a two-parameter grid search, making it feasible to regenerate synthetic scenarios on a daily or weekly cadence as new data arrive [31, 34].

More broadly, the quality evaluation methodology demonstrated here, combining distributional fidelity tests (KS, AD) with temporal structure metrics (ACF-MAE) and reporting standard errors across Monte Carlo ensembles, provides a reusable template for assessing synthetic time series quality in other domains where temporal dependence structures are critical to downstream applications [4].

5.3 Limitations

Several limitations inform the interpretation of these results. First, the model assumes stationarity of the transition matrix and jump hyperparameters across the full in-sample window. The out-of-sample ACF analysis (Figure 4, panel d) provides direct evidence of this limitation: the IS-calibrated (ϵ^*, λ^*) underestimate volatility clustering persistence in the 2025 test window, consistent with the elevated macro uncertainty of that period. Structural breaks in market microstructure, such as those accompanying regulatory changes or the COVID-19 shock, may not be adequately captured by a single static transition matrix.

Second, the quantile-based state partition is fixed at estimation time and does not adapt to evolving volatility regimes. The boundaries are calibrated on the full in-sample distribution and may overweight the central mass relative to the tails during periods of prolonged market stress.

Third, the Single-Index Model extension imposes a linear, single-factor model of cross-sectional dependence. The median IS KS pass rate of 66.7% across 424 assets confirms that this decomposition cannot capture the heterogeneous tail behaviour, skewness, and sector-specific dynamics present in individual equities [15, 39]. However, the architecture is modular: the HMM-WJ generative engine for the market factor is decoupled from the factor model used to propagate paths, and replacing the SIM with a richer specification would not require modifying the underlying Markov chain machinery.

Finally, the out-of-sample evaluation covers a single 249-day window (full calendar year 2025), which limits the statistical power of out-of-sample inference. A longer or rolling evaluation would provide a more robust assessment of generalisation performance.

6 Conclusion

We developed a hybrid hidden Markov model that generates synthetic equity paths whose statistical properties closely match real market data, both in the shape of the return distribution and in the persistence of volatile episodes. The model works by partitioning excess growth rates into quantile-defined regimes and augmenting normal regime-switching with a Poisson jump-duration mechanism that forces the model to linger in extreme states, reproducing the volatility clustering that standard HMMs miss. Applied to SPY over a ten-year training window and validated out-of-sample on the full calendar year 2025, the framework achieves KS and AD pass rates exceeding 95%. The jump mechanism, governed by only two scalar hyperparameters, is the structural feature responsible for generating persistent high-volatility regimes, and the direct frequentist estimation strategy eliminates the computational cost and initialisation sensitivity of the EM algorithm. A Single-Index Model extension scales the approach to a 424-asset universe, providing a baseline for multi-asset synthetic scenario generation.

Several extensions follow naturally. Time-varying transition matrices, estimated via rolling windows or Bayesian online learning, would allow the model to adapt to structural regime shifts. Replacing the single-factor SIM with multi-factor specifications such as the Fama-French three- or five-factor model [15], principal-component-based factor structures, or nonlinear residual generators could substantially improve asset-level distributional fidelity. Adaptive state definitions based on density clustering rather than fixed quantiles could improve tail coverage during crisis periods. Finally, embedding the hybrid HMM as the generative component within a portfolio optimisation loop, where synthetic scenarios drive tail-risk objectives, would provide an end-to-end application of the framework.

Conflict of Interest Statement

The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions

J.V. directed the study. A.A. developed the model and simulation code, conducted the in-sample and out-of-sample analysis, and generated the figures. Both authors edited and reviewed the final manuscript.

Data Availability Statement

The model code, simulation scripts, training and testing data are available under a Massachusetts Institute of Technology (MIT) license from the GitHub repository: https://github.com/varnerlab/HMM-Model-with-Jumps.git

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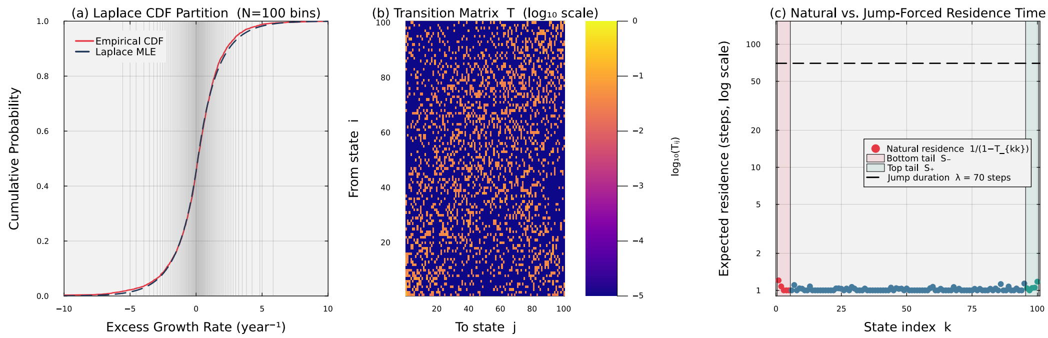

Fig. S1. Fitted model internals for SPY with $N = 100$ states. Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries that define the N hidden states. The close agreement confirms that the Laplace distribution is an accurate marginal model for the excess growth rate data. Panel (b) displays the empirical transition matrix $\mathbf{T}$ in $\log_{10}$ colour scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red shading) and top tail set $\mathcal{S}_+$ (states 96–100, teal shading) highlighted. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering. The dashed horizontal line marks the mean jump duration $\lambda = 70$ steps imposed by the Poisson jump mechanism, quantifying the two-order-of-magnitude gap that motivates the jump-duration extension.

Online Appendix

S1. Fitted model internals

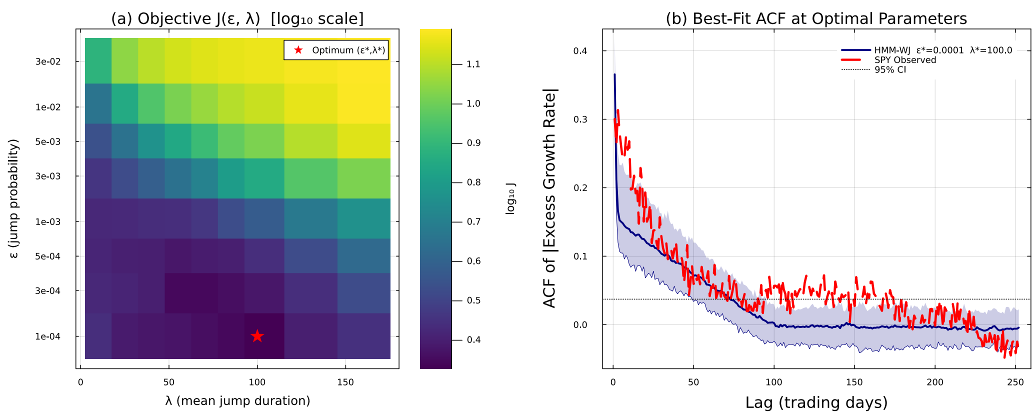

Fig. S2. Hyperparameter grid search over (ϵ, λ) for SPY ($N = 100$, 200 paths per grid point). Panel (a): heatmap of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale over the full search grid; the red star marks the optimal pair (ϵ^*, λ^*). The well-defined interior minimum confirms that the jump hyperparameters are identifiable from data. Panel (b): ACF of $|G_t|$ at the optimal parameters computed from 500 jump-containing paths (10th–90th percentile band shaded; mean curve solid navy) compared with the SPY observed ACF (dashed red) and the 95% significance band (dotted black).

S2. Hyperparameter grid search landscape

Table T1. Sensitivity of model performance to state resolution N. All metrics computed over 1,000 simulated in-sample paths at $\alpha = 0.05$. ACF-MAE = mean absolute error of the ACF of $|G_t|$ over lags 1–252. Standard errors in parentheses: binomial SE for pass rates, bootstrap SE ($B = 500$) for ACF-MAE.

Model	N	**KS pass (%)**	**AD pass (%)**	**ACF-MAE**
HMM-NJ	200	99.6 (0.2)	98.5 (0.4)	0.059
HMM-NJ	150	99.6 (0.2)	98.4 (0.4)	0.059
HMM-NJ	100	99.0 (0.3)	98.8 (0.3)	0.059
HMM-NJ	90	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	60	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	30	98.9 (0.3)	97.9 (0.5)	0.059
HMM-WJ	200	96.0 (0.6)	88.6 (1.0)	0.049
HMM-WJ	150	96.9 (0.5)	88.8 (1.0)	0.050
HMM-WJ	100	96.8 (0.6)	91.7 (0.9)	0.053
HMM-WJ	90	97.4 (0.5)	91.0 (0.9)	0.052
HMM-WJ	60	96.8 (0.6)	92.4 (0.8)	0.054
HMM-WJ	30	98.0 (0.4)	96.8 (0.6)	0.057

S3. State resolution sensitivity

Algorithm 2 Empirical hidden Markov model construction via Laplace partitioning

Require: Price history $P_{1:T}$, Risk-free rate r_f, Number of states N
Ensure: Transition matrix $\mathbf{T}$, Quantile boundaries $\boldsymbol{Q}$

Compute excess growth rates $R_t = \frac{1}{\Delta t}\ln(P_t/P_{t-1}) - r_f$ for all t.
Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation.
Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \ldots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution.
Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$.
Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \ldots, N\}$ based on the boundaries in $\boldsymbol{Q} = \{Q_k\}$.

Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N\times N}$ with zeros.
for $t = 1$ to $T-1$ **do**
    Identify transition $i = s_t$ to $j = s_{t+1}$.
    $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$.
end for
for $i = 1$ to N **do**
    $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:}/\sum_{j=1}^{N}\mathbf{C}_{i,j}$ {Row-wise normalization}
end for
return $\mathbf{T}, \boldsymbol{Q}$

S4. Additional algorithms

The three algorithms below complement Algorithm 1 (main text) by detailing the remaining stages of the computational pipeline: model construction (Algorithm 2), state decoding and continuous rate reconstruction (Algorithm 3), and hyperparameter grid search (Algorithm 4).

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Require: Simulated state sequence $S_{1:M}$, Quantile boundaries $\boldsymbol{Q}$
Ensure: Reconstructed continuous excess growth rates $\hat{R}_{1:M}$

```
Define z_{0.999} = 3.09.
for k = 1 to N do
    Set μ_k = (Q_{k−1} + Q_k)/2.
    Set σ_k = (Q_k − Q_{k−1})/(2z_{0.999}).
end for
Initialize the reconstructed excess growth rate vector Ĝ of length M.
for t = 1 to M do
    Retrieve the simulated state k = S_t.
    Sample the continuous excess growth rate Ĝ_t from the normal distribution N(μ_k, σ_k).
end for
return Ĝ_{1:M}
```

Algorithm 4 Multi-objective grid search for jump hyperparameters

Require: Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$
Require: Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$
Require: Kurtosis penalty weight $w_K = 0.20$
Ensure: Optimal jump hyperparameters (ϵ^*, λ^*)

Calculate observed absolute ACF: $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \ldots, L\}$.
Calculate observed global kurtosis: $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$.
Initialize minimum error $E_{min} \leftarrow \infty$.
for each ϵ in grid space **do**
    for each λ in grid space **do**
        Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$.
        for $p = 1$ to P **do**
            Simulate path using Algorithm 1 to obtain state sequence $S_{1:M}$.
            Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$.
            $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ.
            $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$.
        end for
        Compute ensemble averages: $\overline{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\overline{K}_{sim} = K_{sum}/P$.
        Compute objective: $J = \sum_{\tau=1}^{L} \left(A_{obs}(\tau) - \overline{A}_{sim}(\tau)\right)^2 + w_K \left(K_{obs} - \overline{K}_{sim}\right)^2$.
        if $J < E_{min}$ **then**
            $E_{min} \leftarrow J$
            $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$
        end if
    end for
end for
return ϵ^*, λ^*